

Integer Exponent Laws: Product and Quotient

Answer Key

Grade 8 — Expressions and Equations

Quick Answers

1. 243

2. x^{10}

3. 125

4. y^6

5. 32

6. $3x^3$

7. 1

8. $\frac{1}{64}$

9. —

10. 256

Common wrong answers

5. *If they got 64: multiplied the exponents instead of adding them, giving 2 to the sixth power*

10. *If they got 32768: multiplied the exponents instead of adding them*

How to grade this. Accept any correct route, but the exponent arithmetic must be visible. A student who writes only the final power on problems 5 and 10 has not shown whether the exponents were added or multiplied, which is the whole distinction this sheet tests.

1. Expand and rewrite $3^2 \cdot 3^3$ as one power of 3.

Solution: 3^2 is two factors of 3 and 3^3 is three factors of 3; written together that is $3 \cdot 3 \cdot 3 \cdot 3 \cdot 3$, five factors of 3. Counting factors is exactly what adding exponents does, so $3^2 \cdot 3^3 = \boxed{3^5}$, which equals 243.

2. Write $x^4 \cdot x^6$ as a single power of x .

Solution: Same base, so the factors simply pile up: four of them and then six more. Add the exponents: $x^4 \cdot x^6 = x^{4+6} = \boxed{x^{10}}$.

3. Write $\frac{5^7}{5^4}$ as a single power of 5.

Solution: Expanded, the top holds seven factors of 5 and the bottom holds four. Four pairs cancel, leaving three factors on top — which is what subtracting the exponents records:

$$\frac{5^7}{5^4} = 5^{7-4} = \boxed{5^3}$$

As a number that is 125.

4. Write $\frac{y^9}{y^3}$ as a single power of y .

Solution: Subtract the exponents: $y^{9-3} = \boxed{y^6}$.

5. Simplify $a^3 \cdot a^2$, then evaluate at $a = 2$.

Solution: The bases match, so add the exponents: $a^3 \cdot a^2 = a^{3+2} = a^5$. Substituting $a = 2$ gives $2^5 = \boxed{32}$. Note that multiplying the exponents instead would give a^6 and the value 64 — a different number, which is how you can tell the two laws apart.

6. Simplify $\frac{12x^5}{4x^2}$.

Solution: Handle the coefficient and the variable part separately. The coefficients divide: $12 \div 4 = 3$. The powers of x follow the quotient law: $x^{5-2} = x^3$. Together:

$$\frac{12x^5}{4x^2} = \boxed{3x^3}$$

7. Read $\frac{6^4}{6^4}$ two ways and say what 6^0 must equal.

Solution: (a) By cancelling: the top and the bottom are the same number, so the quotient is 1. (b) By the quotient law: $6^{4-4} = 6^0$. Both readings describe the same expression, so 6^0 and 1 are the same number:

$$6^0 = \boxed{1}$$

This is why any nonzero base to the zero power is defined as one — it is forced by the quotient law, not an arbitrary rule.

8. Apply the quotient law to $\frac{4^2}{4^5}$.

Solution: Subtract as usual, even though the answer is negative: $4^{2-5} = 4^{-3}$. Expanding shows the same thing: two factors on top cancel two of the five below, leaving three factors of 4 in the denominator.

$$\frac{4^2}{4^5} = \boxed{4^{-3}} = \frac{1}{4^3} = \frac{1}{64}$$

A negative exponent is a location (denominator), never a negative number.

9. Compare $3^4 \cdot 3^2$ with $\frac{3^7}{3^1}$.

Solution: Simplify each side to a single power of 3 before comparing. Left: add the exponents, $3^{4+2} = 3^6$. Right: subtract the exponents, $3^{7-1} = 3^6$. Same base, same exponent, so the two expressions are the same number (729), and the correct symbol is $\boxed{=}$.

10. Jae's error: $2^5 \cdot 2^3$ answered as 32768.

Solution: Jae multiplied the exponents. The product of powers law adds them, because the two groups of factors are being placed side by side, not nested: $2^5 \cdot 2^3 = 2^{5+3} = 2^8$. Evaluating, $2^8 = \boxed{256}$. Jae's route computed 2^{15} instead, which is more than a hundred times too large — a useful reasonableness check on its own.